MILITARY SPECIFICATION
INITIATOR, PROPELLANT ACTUATED, DELAY, M113 ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Propellant Actuated, Delay, M113, Assembly loaded with 2R pellets, an M42 primer and a delay element for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-H-25579	- Hose Assembly Tetrafluoroethylene, High Temperature Power Plant, Aircraft

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

DRAWINGS

Munitions Command

F11731474	- Initiator, Propellant Actuated, Delay, M113, Assembly

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2.1 Drawings (Cont'd)

C9252573	- Carton, Packing, Ammunition for Initiator, M113
C9252574	- Box, Packing, Ammunition for Initiator, M113
IEL 11731474	- Inspection Equipment List for Initiator, Propellant Actuated, Delay, M113 Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Breakaway torque.- Each threaded joint of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.3 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.4 Functional.

3.4.1 Ballistics.-Initiators loaded with acceptable pyrotechnic components (2R pellets, M42 primer and delay element) and temperature conditioned shall produce pressure values specified in Table I. Pressure values shall be recorded at the end of 15 feet of aircraft hose, size No. 4 as specified in MIL-H-25579.

TABLE I

Temperature	Pressure (PSI) Min.	Max.	Ignition Delay, Max. (see 6.3)
-65° to -70° F	1000	2300	0.450 Sec, Max.
65° to 75° F	1000	2300	0.300 ± 0.050 Sec.
200° to 205° F	1000	2300	0.150 Sec., Min.

3.4.1.1 Initiation.- The initiator shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, size No. 4, in accordance with MIL-H-25579 at 70° ± 5° F.

3.4.2 The initiator shall not fail to fire.

3.4.3 The initiator shall function without mechanical failure (see 6.2).

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3.5 Atmospheric seal.- The assembled initiator, less the firing mechanism shall withstand 14 PSI, internal, external pressure differential leak rate of 1 X 10^{-4} cubic centimeters per second.

3.6 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.- The part shall not have a burr which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

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4.2.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with acceptable explosive and pyrothecnic components.

4.2.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Table II. All non-conforming material shall be rejected. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table II. Four power magnification may be used for inspection of such defects as burr and foreign matter.

TABLE II

		Major		Minor	
Lot Size	Sample Size	Accept	Reject	Accept	Reject
1 - 300	100% examination	-	-	-	-
301 - 800	50	0	1	1	2
801 -1800	90	1	2	2	3

4.2.2.1 Assembly, Drawing F11731474, less all drawings except Drawings C11738410, B11731626, D11731622 and MS 28775-009.

Defect	Method of Inspection
CRITICAL	
1. Firing pin protrusion	Gage
2. Shear pin not engaged in firing pin	Gage
3. Striker hole in initiator cap obstructed	Visual
MAJOR	
101. O-ring not lubricated	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.2 Assembly, Drawing F11731474.

Defect	Method of Inspection
CRITICAL	
1. Any part missing except as otherwise classified	X-ray 1/ and Visual
2. Improper assembly	X-ray 1/ and Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
201. Thread sealing compound missing	Visual
202. Shipping cap missing	Visual
203. Protective finish inadequate or defective	Visual

1/ The assembly shall be x-rayed with the long axis of the initiator perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.2.2.3 Carton, Packing, Ammunition for Initiator, M113, Dwg C9252573.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Cut or damaged through all impregnated layers	Visual
203. Marking missing, incorrect or illegible	Visual
204. Tape incomplete or badly wrinkled	Visual
205. Length of tear tab, min.	Visual
206. Color of tape incorrect	Visual

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4.2.2.4 Box, Packing, Drawing C9252574.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3:

4.2.3.1 Breakaway torque.- Each initiator shall be subjected to the breakaway torque application. Failure of any initiator assembly to comply with the requirements of 3.2 shall be cause for rejection of the assembly.

4.2.3.2 Residual magnetism.- Each initiator shall be subjected to the test for excessive residual magnetism. Failure of any initiator to comply with the requirements of 3.3 shall be cause for rejection of the initiator.

4.2.3.3 Atmospheric seal.- Each initiator shall be subjected to an atmospheric seal test. Failure of any initiator to comply with the requirements of 3.5 shall be cause for rejection of the initiator.

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4.2.3.4 Functional.- The following tests shall be performed by the Government:

4.2.3.4.1 Ballistics.- Twenty-one units shall be tested to determine compliance with the requirements of 3.4.1, 3.4.2 and 3.4.3. The units shall be temperature conditioned as provided in Table III and 4.3.3.1.1. Failure of any unit to comply with the requirements of 3.4.1, 3.4.2 and 3.4.3 shall be cause for rejection of the lot.

TABLE III

Temperature	Sample Size
-65° to -70°F	7
65° to 75°F	7
200° to 205°F	7

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL 10520555.

4.3 Test methods and procedures

4.3.1 Breakaway torque.- A torque wrench set at the torque requirement specified in 3.2 shall be used in conjunction with the fixture to determine the adequacy of the joint. The torques shall be applied (1) between the female body and male body and (2) between the male or female body and the body and (3) between the body and the housing.

4.3.2 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects, by placing the initiator 6 inches from and at the same height as the compass in a north-south horizontal position (housing end at south) with the compass set in an east-west heading. The initiator shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the initiator.

4.3.3 Atmospheric seal.- The method and procedure for performing the atmospheric seal test shall be as approved by the Government.

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4.3.4 Functional.

4.3.4.1 Ballistics.- The test sample shall be rigidly supported in the fixture and shall be tested within five (5) minutes after its removal from the conditioning chamber. When this time limit is exceeded the sample shall be reconditioned in accordance with 4.3.3.1.1.

4.3.4.1.1 Temperature conditioning.- A standard M113 Initiator loaded with dummy 2R pellets and a dummy M42 primer and delay element containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C9252573 and C9252574.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure time record.

Custodian:

Army - MU

Preparing Activity:

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